

KARTHICK B

AI Engineer

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Professional Summary

Motivated **AI Engineer** and B.Tech CSE graduate specializing in **Artificial Intelligence & Data Science**, with hands-on experience in **machine learning, deep learning, computer vision, NLP, and Python development**. Built and deployed real-world AI projects including a **YOLOv8-based cricket analytics system**, a **RAG chatbot** using FAISS and Groq LLaMA, and a **Tamil Nadu government scheme finder**. Strong foundations in **model training, ML pipelines, data pipelines, and API-based model deployment** using FastAPI and Streamlit. Seeking a Machine Learning Engineer or AI Engineer role to build and deploy intelligent systems at scale.

Education

Dr. M.G.R. University, B.Tech – CSE (Artificial Intelligence & Data Science) June 2021 – May 2025
Chennai, Tamil Nadu CGPA: 8.1 / 10

Maharishi Higher Secondary School – HSC Mar 2021
Tiruvannamalai, Tamil Nadu Percentage: 90%

Technical Skills

- **Languages:** Python, SQL
- **AI / ML:** Scikit-learn, YOLOv8, OpenCV, FAISS, Sentence Transformers, LangChain, Hugging Face, Groq LLaMA, NLP
- **Frameworks:** TensorFlow, PyTorch
- **Web & Deployment:** FastAPI, Streamlit, Hugging Face Spaces
- **Data & Visualization:** NumPy, Pandas, Matplotlib
- **Tools:** Git, GitHub, Google Colab, VS Code

Experience

Besant Technologies – Chennai, Tamil Nadu Dec 2024 – Mar 2025
Software Development Intern

- Completed hands-on training in **Core Java, MySQL, and Manual Testing**, covering SDLC concepts and structured test case design.
- Wrote and executed test cases for web application modules; practiced defect tracking and SQL-based backend validation.
- Gained practical exposure to **OOP design patterns, JDBC connectivity, and database operations** through real-world project exercises.

Projects

RAG Chatbot (Python, FAISS, Groq LLaMA, FastAPI) 📄 Rag_chatbot | 🌐 Live

- Built a **Retrieval-Augmented Generation (RAG)** chatbot that answers questions from a custom knowledge base using **FAISS vector search** and **Sentence Transformers (all-MiniLM-L6-v2)** for semantic retrieval.
- Integrated **Groq LLaMA 3.3 70B** as the LLM backend for fast, context-aware response generation with top-3 chunk retrieval.
- Developed a **FastAPI** backend with auto Swagger docs and a dark-themed **HTML/JS** chat frontend; **deployed live on Hugging Face Spaces**.

AI Cricket Analytics System (Python, YOLOv8, OpenCV, Streamlit) 📄 Cricket-CV

- Built an AI-powered cricket analytics web app with **real-time ball detection** using YOLOv8 and OpenCV, enabling frame-level tracking at match speed.
- Implemented **ball trajectory tracking** and **dynamic wide-ball detection** logic for automated match event recognition.
- Developed a **Streamlit-based dashboard** featuring match analytics, live webcam detection, and output video generation for end-to-end demo deployment.

TN Scheme Finder (Python, NLP, Streamlit, Web Scraping) 📄 TN-SchemeFinder

- Developed a web application that helps citizens **discover Tamil Nadu government schemes** based on eligibility criteria using NLP-based matching.
- Implemented **keyword-based and semantic search** to retrieve relevant scheme information from a structured knowledge base.
- Built an interactive **Streamlit UI** enabling users to filter schemes by category, age group, and occupation for easy access.